

Git Branching and Merging Assignment

Objective

Create separate branches for favorite foods and favorite movies, merge them into the main branch, resolve merge conflicts, and push the final code to GitHub.

Step 1: Initialize Git Repository

```
git init
git remote add origin https://github.com/Abhinavj12/Movie_Food
git remote -v
```

Result:

- Local Git repository initialized.
 - GitHub repository connected as remote origin.
-

Step 2: Add Initial File and Commit

```
git add favourite.txt
git commit -m "This is my First commit"
```

Result:

- Initial commit created on the master branch.
-

Step 3: Rename Master Branch to Main

```
git branch -m main
git branch
```

Result:

- Default branch renamed from master to main.
-

Step 4: Push Main Branch to GitHub

```
git push origin main
```

Result:

- Main branch successfully pushed to GitHub.
-

Step 5: Create and Work on Foods Branch

```
git branch foods  
git switch foods
```

Added favorite food items to favourite.txt and committed changes:

```
git add favourite.txt  
git commit -m "Adding favourite food content in food branch"
```

Result:

- Foods branch created and food-related content added.
-

Step 6: Create and Work on Movies Branch

```
git switch main  
git branch movies  
git switch movies
```

Added favorite movie items to favourite.txt and committed changes:

```
git add favourite.txt  
git commit -m "This is my 2nd commit, by adding fav movies content in movies  
branch"
```

Result:

- Movies branch created and movie-related content added.
-

Step 7: Push Both Branches to GitHub

```
git push origin foods  
git push origin movies
```

Result:

- Foods and Movies branches successfully pushed to GitHub.
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Step 8: Merge Foods Branch into Main

```
git switch main  
git merge foods
```

Result:

- Foods branch merged successfully using fast-forward merge.
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Step 9: Merge Movies Branch into Main

```
git merge movies
```

Result:

- Merge conflict occurred because both branches modified the same file (favourite.txt).

Error:

```
CONFLICT (content): Merge conflict in favourite.txt  
Automatic merge failed; fix conflicts and then commit the result.
```

Step 10: Resolve Merge Conflict

Edited favourite.txt manually and combined both favorite foods and favorite movies into a single file.

Then executed:

```
git add favourite.txt  
git commit -m "Resolved merge conflict and combined foods and movies"
```

Result:

- Merge conflict resolved successfully.
-

Step 11: Push Final Main Branch

```
git push origin main
```

Result:

- Updated main branch pushed to GitHub successfully.
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Final Git Log

```
git log --oneline --graph --all
```

Output:

```
*    59a6aa8 Resolved merge conflict and combined foods and movies
|
| * d6444fe This is my 2nd commit, by adding fav movies content in movies
branch
* | 8b07360 Adding favourite food content in food branch
|/
* 53d5faa This is my First commit
```

Conclusion

This assignment demonstrated Git branching, committing changes, pushing branches to GitHub, merging branches, handling merge conflicts, and maintaining version control. The final main branch contains both favorite food and favorite movie entries after successfully resolving the merge conflict.